

Theme 2.1 Productivity and National Economic Performance — Edexcel B

1. Measuring National Income — GDP

Gross Domestic Product (GDP): The total monetary value of all final goods and services produced within a country's borders in a given time period (usually a year or quarter). It is the most widely used measure of economic output and national income.

1.1 Three Methods of Measuring GDP

All three approaches should give the same figure (the national income identity):

Method	What It Measures	Calculation
Output (Production)	Value added by each sector of the economy	Sum of value added in agriculture + manufacturing + services + ...
Income	Total incomes earned by factors of production	Wages + rent + interest + profits
Expenditure	Total spending on final goods and services	$C + I + G + (X - M)$

$$\text{GDP (Expenditure)} = C + I + G + (X - M)$$

Where: C = consumer spending, I = investment, G = government spending, X = exports, M = imports.

1.2 Real vs Nominal GDP

- **Nominal (money) GDP:** Measured at current prices. Changes in nominal GDP may reflect price changes (inflation) rather than real changes in output.
- **Real GDP:** Adjusted for inflation (measured at constant/base-year prices). Shows the actual change in the volume of output. This is the more meaningful measure.

Real GDP = Nominal GDP adjusted for inflation using a GDP deflator

1.3 GDP per Capita

GDP per capita = $\text{GDP} \div \text{Population}$

A better indicator of average living standards than total GDP because it accounts for population size. A country with high GDP but a very large population may have low GDP per capita (e.g., India).

1.4 Limitations of GDP as a Measure of Living Standards

- **Distribution of income:** GDP per capita is an average — it masks inequality. A country could have high GDP per capita but most wealth concentrated among a few.
- **The informal/shadow economy:** GDP does not capture unrecorded economic activity (cash-in-hand work, illegal trade, subsistence farming). This is larger in developing countries.
- **Non-marketed output:** Unpaid work such as housework, childcare, and voluntary work is excluded.
- **Quality of life factors:** GDP ignores leisure time, environmental quality, health, education quality, personal freedom, and happiness.
- **Negative externalities:** GDP counts all output, including production that causes pollution and environmental destruction. "Bads" like oil spill clean-ups increase GDP.
- **Composition of output:** GDP does not distinguish between spending on weapons and spending on hospitals.
- **Purchasing Power Parity (PPP):** When comparing GDP across countries, exchange rate differences can distort comparisons. PPP-adjusted figures give a more accurate picture of relative living standards.

1.5 Alternative Measures of Living Standards

Measure	What It Includes	Advantage over GDP
HDI (Human Development Index)	Life expectancy, education (mean/expected years of schooling), GNI per capita (PPP)	Broader measure of development, not just income
GNI (Gross National Income)	GDP + net income from abroad	Captures income earned by nationals overseas
Green GDP	GDP adjusted for environmental degradation	Accounts for sustainability
Genuine Progress Indicator	GDP adjusted for inequality, environmental costs, non-market activity	More holistic welfare measure

Case Study: China's GDP Growth and Living Standards

China's GDP grew at an average of around 10% per year from 1980 to 2015, lifting hundreds of millions out of poverty. However, GDP growth overstated improvements in living standards for several reasons: severe air pollution in cities like Beijing imposed enormous health costs not captured by GDP; the Gini coefficient rose from 0.30 in 1980 to over 0.47 by 2012 (highly unequal); and much growth came from environmentally destructive heavy industry. China's HDI (0.788 in 2022) ranks significantly lower than its GDP ranking would suggest, illustrating GDP's limitations.

2. Inflation

Inflation: A sustained increase in the general price level over a period of time — meaning money loses purchasing power. It is measured as the annual percentage change in a price index.

Deflation: A sustained decrease in the general price level (negative inflation rate).

Disinflation: A fall in the rate of inflation (prices still rising, but more slowly).

2.1 Measuring Inflation — CPI and RPI

Feature	CPI (Consumer Price Index)	RPI (Retail Price Index)
Used by	Government's target measure; used by the Bank of England	Index-linked gilts, some pensions, rail fare increases
Includes housing costs?	Partially (CPIH includes owner-occupier housing costs)	Yes — includes mortgage interest payments
Coverage	Broader population	Excludes top 4% earners and pensioners relying on state benefits
Typical value	Usually lower than RPI	Usually higher than CPI

Both use a weighted basket of around 700 goods and services, updated annually to reflect changing spending patterns. The Living Costs and Food Survey samples around 5,000 households to determine weights. Items are added (e.g., oat milk, smartwatches) and removed (e.g., DVD players) each year to keep the basket representative.

2.2 Causes of Inflation

Demand-Pull Inflation

Caused by aggregate demand (AD) increasing faster than aggregate supply (AS) — "too much money chasing too few goods." When the economy is near or at full capacity, increased spending leads to price rises rather than output increases.

Causes of rising AD: Consumer confidence, lower interest rates, tax cuts, rising asset prices (wealth effect), increased government spending, weaker exchange rate (boosting exports).

Cost-Push Inflation

Caused by increases in the costs of production — firms pass on higher costs as higher prices. This shifts the SRAS curve to the left.

Causes: Rising oil/energy prices, higher wages (above productivity growth), weaker exchange rate (making imports more expensive), higher commodity prices, increased indirect taxes, supply chain

disruptions.

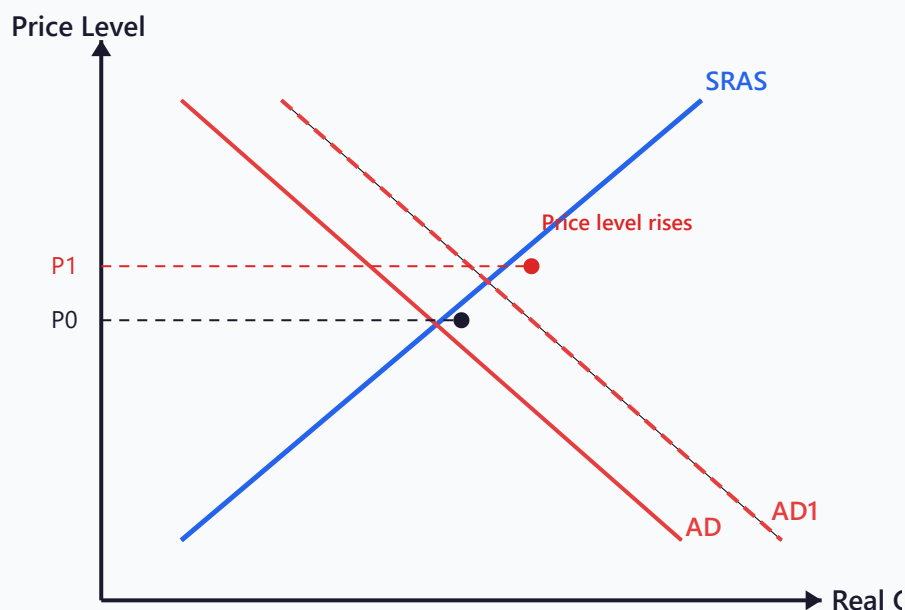


Figure 1: Demand-pull inflation — AD shifts right, price level rises from P0 to P1.

2.3 Consequences of Inflation

Effect	Explanation
Reduced purchasing power	Money buys less — real incomes fall if wages do not keep pace with inflation.
Menu costs	Firms must update price lists, catalogues, and vending machines more frequently.
Shoe-leather costs	People spend more time and effort managing their money to minimise the erosion of real value.
Redistribution of income	Borrowers benefit (repay loans in devalued money); savers and those on fixed incomes lose.

Reduced international competitiveness	Higher domestic prices make exports more expensive and imports cheaper, worsening the trade balance.
Uncertainty	High/volatile inflation makes it harder for firms to plan investment — may reduce long-run growth.
Fiscal drag	If tax thresholds are not adjusted for inflation, more people are dragged into higher tax brackets.

2.4 Consequences of Deflation

Deflation can be even more dangerous than inflation:

- **Delayed spending:** Consumers postpone purchases expecting further price falls — reduces AD further (deflationary spiral).
- **Rising real debt burden:** The real value of debt increases, causing defaults and financial instability.
- **Falling profits and wages:** Firms see revenue fall but may have fixed nominal costs — squeezing profits and leading to redundancies.
- **Monetary policy ineffectiveness:** Interest rates cannot go below zero (the zero lower bound), making it hard to stimulate the economy.

Case Study: UK Cost-of-Living Crisis 2022-23

UK CPI inflation peaked at 11.1% in October 2022, the highest in 41 years. The primary drivers were cost-push: energy prices surged following Russia's invasion of Ukraine (wholesale gas prices rose over 400%), global supply chain disruptions persisted post-COVID, food prices rose sharply due to higher fertiliser and transport costs, and a weakened pound made imports more expensive. Real wages fell for many months — average earnings growth of around 6% was far below inflation — causing the sharpest fall in living standards since records began. The Bank of England raised the base rate from 0.1% to 5.25% to combat inflation.

3. Unemployment

Unemployment: When people who are willing and able to work, and are actively seeking employment, cannot find a job. The UK uses two measures: the **Claimant Count** (those claiming unemployment-related benefits) and the **ILO/Labour Force Survey** measure (those without a job who have sought work in the last 4 weeks and are available to start within 2 weeks).

3.1 Measuring Unemployment

Measure	Method	Strengths	Weaknesses
Claimant Count	Number claiming JSA or UC (searching for work element)	Monthly, cheap, accurate count	Excludes those not eligible or not claiming; changes with benefit rules
ILO/LFS	Quarterly survey of 40,000 households	Internationally comparable; counts hidden unemployment	Sample size — subject to sampling error; published with a lag

3.2 Types and Causes of Unemployment

Cyclical (Demand-Deficient) Unemployment

Caused by a fall in aggregate demand during a recession or downturn. When AD falls, firms reduce output and lay off workers. This is the most significant type during recessions and is directly linked to the economic cycle.

Structural Unemployment

Caused by a long-term decline in certain industries due to changes in technology, international competition, or consumer demand. Workers' skills no longer match the jobs available — there is a **mismatch** between the skills of the unemployed and the requirements of available vacancies. Can be geographical (jobs in different regions) or occupational (different skills needed).

Frictional Unemployment

Short-term unemployment that occurs as people move between jobs. Inevitable in a dynamic economy — people leave one job and take time to find the next. Generally considered non-harmful and even

desirable (better job matching).

Seasonal Unemployment

Unemployment that occurs at certain times of year due to seasonal patterns in demand (e.g., tourism workers in winter, agricultural workers between harvests).

Real Wage (Classical) Unemployment

Occurs when wages are held above the market-clearing level (e.g., by minimum wage laws, trade union power), creating excess supply of labour. At the above-equilibrium wage, quantity of labour supplied exceeds quantity demanded.

3.3 Natural Rate of Unemployment (NRU)

Natural rate of unemployment: The level of unemployment that exists when the economy is at its long-run equilibrium (full employment) — consisting of frictional, structural, and seasonal unemployment. At the NRU, there is no cyclical unemployment, and inflation is stable. Also known as the **NAIRU** (Non-Accelerating Inflation Rate of Unemployment).

3.4 Consequences of Unemployment

For Individuals	For the Economy	For the Government
Loss of income and lower living standards	Lost output — economy inside its PPF	Higher spending on benefits (JSA, UC)
De-skilling — skills atrophy over time	Hysteresis — long-term unemployment makes future employment harder	Lower tax revenue (income tax, VAT)
Mental health issues, stress, loss of self-esteem	Reduced consumer spending and business confidence	Higher budget deficit
Social problems — crime, family breakdown	Regional disparities widen	Increased demand for public services

Case Study: UK Post-Industrial Structural Unemployment

The decline of coal mining, steel production, and shipbuilding in the 1980s left communities in South Wales, Northern England, and Central Scotland with severe structural unemployment. Workers had skills specific to heavy industry that were not transferable to the growing service sector concentrated in the South East. Decades later, these areas still have higher unemployment rates and lower average incomes than the national average — a clear example of hysteresis and the geographical immobility of labour. Government responses have included Enterprise Zones, regional development agencies, and retraining programmes, with mixed success.

4. Balance of Payments

Balance of payments: A record of all financial transactions between residents of a country and the rest of the world over a period of time. It always balances overall (surpluses in one account are offset by deficits in another).

4.1 Structure

Account	What It Records	Key Components
Current Account	Trade in goods and services, plus income and transfers	Trade balance (goods), trade in services, primary income (investment income), secondary income (transfers, e.g., foreign aid, contributions to international organisations)
Capital Account	Capital transfers and non-produced, non-financial assets	Debt forgiveness, sale of patents, EU structural funds (small account)
Financial Account	Investment flows and changes in reserves	Foreign Direct Investment (FDI), portfolio investment, reserve assets, other investment

$$\text{Current Account} + \text{Capital Account} + \text{Financial Account} = 0$$

4.2 Current Account Deficit

The UK has run a persistent current account deficit since the mid-1980s (typically around 2-4% of GDP in recent years). This means the UK imports more goods and services than it exports, and pays out more in investment income and transfers than it receives.

Causes of the UK's Current Account Deficit

- **High consumer spending on imports:** UK consumers have a high marginal propensity to import — strong demand for foreign goods (cars, electronics, energy).
- **Decline of manufacturing:** De-industrialisation since the 1980s reduced the UK's goods export capacity.
- **Strong pound (historically):** Made UK exports expensive and imports cheap.
- **Low savings rate:** UK consumers save less and spend more, including on imports.
- **Primary income deficit:** Foreign ownership of UK assets means investment income flows out.

Consequences of a Persistent Deficit

- Must be financed by capital inflows (selling assets to foreigners, borrowing).
- Increases foreign ownership of domestic assets — income flows out in future.
- Could lead to currency depreciation if financed by short-term "hot money" that can withdraw suddenly.
- May not be a problem if deficit reflects high investment inflows (FDI creates jobs and growth).

Evaluation: How Significant Is a Current Account Deficit?

- A deficit financed by FDI may actually be beneficial — foreign firms create jobs and transfer technology.
- The US has run a deficit for decades without crisis — the dollar's reserve currency status helps.
- A deficit reflects high domestic demand — may indicate a strong economy.
- However, a deficit financed by short-term capital flows is vulnerable to sudden stops.
- Persistent deficits may indicate a structural competitiveness problem.

5. The Economic (Business) Cycle

Economic cycle: The regular fluctuations in real GDP around the long-run trend rate of growth. The economy moves through phases of boom, slowdown, recession, and recovery.

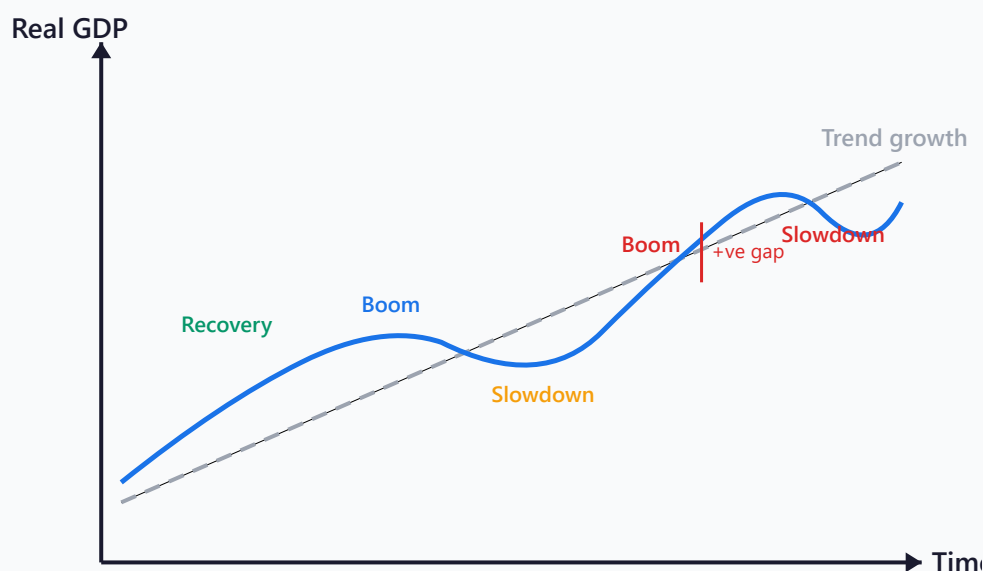


Figure 2: The economic cycle — real GDP fluctuates around the long-run trend rate of growth.

5.1 Phases of the Cycle

Phase	Characteristics
Boom	High GDP growth, low unemployment, rising wages, high consumer/business confidence, rising inflation, positive output gap. Risk of economy overheating.
Slowdown	Growth decelerates, confidence falls, some firms fail, unemployment begins rising.
Recession	Technically defined as two consecutive quarters of negative GDP growth. Rising unemployment, falling business investment, falling consumer spending, falling inflation or deflation, negative output gap.

Recovery

GDP begins to grow again, unemployment starts falling, confidence returns, investment picks up.

5.2 The Output Gap

Output gap: The difference between actual GDP and potential GDP (the maximum sustainable output of the economy).

Positive output gap: Actual GDP > potential GDP — economy overheating, inflationary pressures.

Negative output gap: Actual GDP < potential GDP — spare capacity, higher unemployment, deflationary pressures.

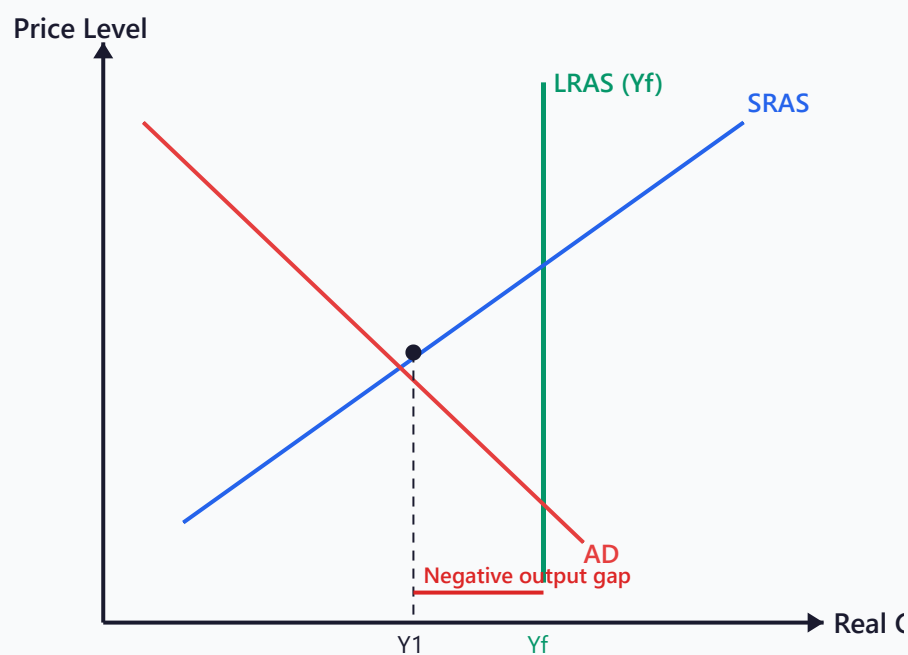


Figure 3: A negative output gap — actual output Y_1 is below potential output Y_f , indicating spare capacity and unemployment.

Case Study: The 2008-09 UK Recession

The global financial crisis triggered the UK's deepest recession since WWII. GDP fell by 6.3% from peak to trough. Unemployment rose from around 5% to over 8% (peaking at 8.5% in late 2011). The banking system was on the verge of collapse — the government had to nationalise Northern

Rock and inject £45.5 billion into RBS and over £20 billion into Lloyds. The Bank of England cut rates to 0.5% and introduced quantitative easing. The fiscal deficit reached over 10% of GDP. The recovery was slow — real GDP did not return to its pre-crisis peak until 2013, and the decade of low productivity growth that followed is called the "lost decade."

6. Productive Efficiency

Spec 2.3. National economic performance is not only a set of headline numbers — it is built out of how efficiently individual firms turn inputs into output. This section moves from the macro picture to the factory floor and back again.

6.1 Productivity

Productivity: output per unit of input in a given time period. Labour productivity is usually measured as output per worker or output per hour worked.

$$\text{Labour productivity} = \text{total output} \div \text{number of workers (or hours worked)}$$

Worked example. A plant produces 96,000 units a year with 60 workers. Labour productivity = $96,000 \div 60 = \mathbf{1,600 \text{ units per worker per year}}$. After investing in new machinery, output rises to 115,200 units with 64 workers: $115,200 \div 64 = \mathbf{1,800 \text{ units per worker}}$. Percentage change in productivity = $(1,800 - 1,600) \div 1,600 \times 100 = \mathbf{12.5\%}$. Note that output rose 20% while productivity rose only 12.5% — part of the extra output came from hiring, not from working more efficiently. Separating those two is exactly what a data-response question is testing.

Factors influencing productivity

- **Capital per worker** — better machinery, software and premises. This is the main long-run driver, and it requires investment, which requires finance.
- **Skills, education and training** — human capital. Slow to build, easily lost if trained workers leave.
- **Management quality** — how work is organised, scheduled and measured. Often the cheapest source of gains.

- **Motivation** — pay structure, job design, security.
- **Technology and process innovation** — automation, better information systems.

Why productivity matters beyond the firm

- **Productivity and competitiveness:** higher output per worker lowers unit labour cost, so the firm can undercut rivals or hold price and earn a wider margin. The same logic applies to a whole economy's exports.
- **Productivity and wages:** over the long run, real wages can only rise sustainably if output per worker rises. A pay rise not matched by productivity raises unit costs and feeds through to prices.
- **Productivity and economic growth:** an economy grows either by using more inputs or by using them better. Once the workforce stops growing, productivity growth is the only source of rising GDP per head — which links this section directly back to the living-standards discussion in section 1.

Labour-intensive and capital-intensive production

	Labour intensive	Capital intensive
Cost structure	Mostly variable cost — wages rise and fall with output	Mostly fixed cost — machinery must be paid for whether it runs or not
Flexibility	Output can be scaled up or down quickly	Hard to scale down; high break-even output
Best suited to	Bespoke, variable or service work; low labour-cost locations	Standardised, high-volume output
Risk in a downturn	Lower — costs fall with sales	Higher — fixed costs continue while revenue falls

6.2 Capacity Utilisation

Capacity utilisation: the proportion of a firm's maximum possible output that it is currently producing.

$$\text{Capacity utilisation (\%)} = \text{current output} \div \text{maximum possible output} \times 100$$

Full capacity means output is at the maximum the existing plant and workforce can produce. **Spare capacity** is the gap between that maximum and actual output.

Worked example. A factory can make 25,000 units a month and is currently making 17,500.
Capacity utilisation = $17,500 \div 25,000 \times 100 = 70\%$.

Fixed costs are £150,000 a month, so fixed cost per unit = $£150,000 \div 17,500 = \textbf{£8.57}$ (to 2 d.p.).
If orders rise to 22,500 units, utilisation = $22,500 \div 25,000 \times 100 = 90\%$, and fixed cost per unit falls to $£150,000 \div 22,500 = \textbf{£6.67}$.

Raising utilisation from 70% to 90% cut fixed cost per unit by $£8.57 - £6.67 = \textbf{£1.90}$, a fall of $1.90 \div 8.57 \times 100 = \textbf{22.2\%}$ — without buying a single new machine. That is the whole argument for managing capacity.

Under-utilisation (low %)	Over-utilisation (near or at 100%)
Fixed costs spread over few units, so average cost is high	No slack to handle a breakdown, a rush order or an absence
Idle machinery and staff; morale and job-security worries	Machines run without maintenance windows, so failures rise
Signals to rivals and investors that demand is weak	Overtime and rushed work raise costs and lower quality
But: room to accept a large new order immediately	Staff burnout, higher turnover, missed delivery dates

Most firms therefore aim for high but not maximum utilisation — typically somewhere around 85–90% — leaving a deliberate margin for maintenance and for demand that arrives unannounced.

Ways of improving capacity utilisation

- **Raise demand** — promotion, new markets, off-peak pricing to fill quiet periods.
- **Subcontract in** — produce for another firm under its brand, filling spare capacity with someone else's orders.
- **Rationalise** — cut the maximum: close a site, sell equipment, reduce hours. This raises utilisation by shrinking the denominator, and it is largely irreversible.
- **Redeploy** — move staff and machinery to product lines where demand is stronger.

6.3 Efficiency and Competitiveness Using Lean Production

Lean production is a set of methods aimed at producing more with less — less stock, less waste, less time, less rework.

Method	What it does	What it costs or risks
Quality control	Inspects finished output and rejects defects at the end of the line	Defective units are already made, so the materials and labour in them are wasted
Quality assurance	Builds checks into every stage so defects are caught where they occur	Needs training and worker responsibility at every stage
Total Quality Management (TQM)	Makes quality the responsibility of every employee, with the internal "next stage" treated as a customer; aims at zero defects	A culture change, not a procedure — slow to embed and easily lost when staff turn over
Kaizen (continuous improvement)	Many small improvements suggested by the people doing the work, applied continuously rather than in big projects	Requires genuine willingness to act on suggestions; gains are incremental, so it will not rescue a failing product
Just in time (JIT)	Stock arrives as it is needed rather than being held; frees warehouse space and working capital	No buffer, so one late delivery, strike or border delay stops production entirely

Evaluation: lean production and competitive advantage

- **The cost case:** less stock means less working capital tied up and lower storage and insurance costs; fewer defects means less wasted material and less rework.
- **The revenue case:** consistent quality supports a price premium and repeat purchase; shorter development lead times let the firm reach the market before rivals, which matters most where products date quickly.
- **The resilience case against:** JIT converts a cost saving into a supply-chain risk. A firm holding no buffer stock saves money in every normal year and loses production in the abnormal one.
- **It depends on** supplier reliability, the distance and complexity of the supply chain, how predictable demand is, and whether the workforce is stable enough for TQM and Kaizen to take hold.
- **Counter-argument:** lean methods are widely known and widely copied, so they may be necessary to compete without being sufficient to win. Sustained advantage usually needs something rivals cannot imitate.

6.4 Impact on Costs and Sales Revenue

- **Average costs fall** for two separate reasons: higher utilisation spreads fixed costs over more units, and less waste lowers variable cost per unit. Keep the two mechanisms apart in an answer — examiners reward the distinction.
- **Minimising waste of resources** — scrap, rework, excess stock, unnecessary movement and waiting time are all inputs paid for that produce nothing saleable.
- **Short product development lead times** — getting from idea to sale faster means responding to a demand shift before rivals, charging a higher launch price, and recovering development costs sooner.

Exam Tip: When discussing macroeconomic objectives, always acknowledge potential conflicts — reducing unemployment by raising demand may push up inflation; boosting growth may worsen the current account as imports are sucked in. Productivity is the one variable that can relieve several conflicts at once, because it allows output and real wages to rise without adding to cost pressure. The best answers explain when a trade-off binds and when it does not.

